DIGITAL CLOCK USING FPGA

Indian Institute of Space Science and Technology

B. Tech Project

AV241: Digital Electronics and VLSI Lab

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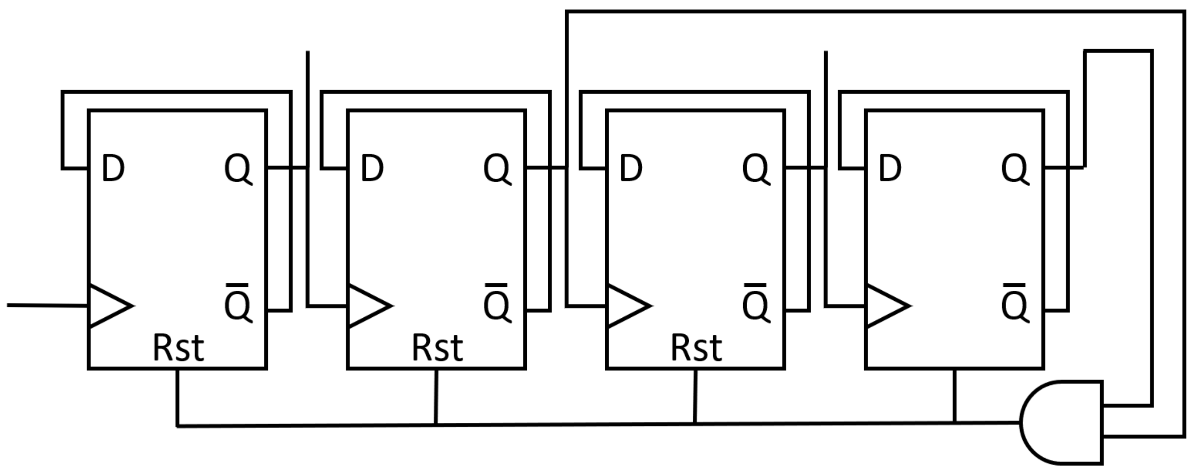
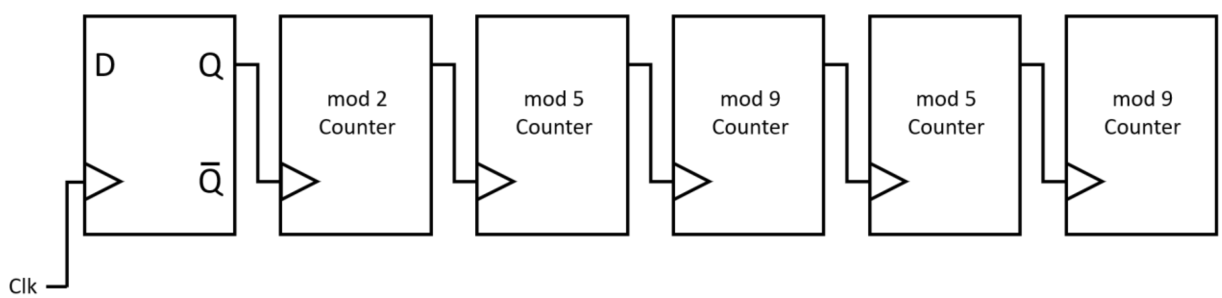
Aim / Problem Statement:
To make a digital clock that displays time in hours, minutes and seconds and has an option of setting time.

Approach:
The idea is to make the digital clock using D-Flip Flop based counters. We will use mod-9 counters, mod-5 counters and D-Flip Flops. Below is the diagram of a mod-9 counter made using D-Flip Flop.

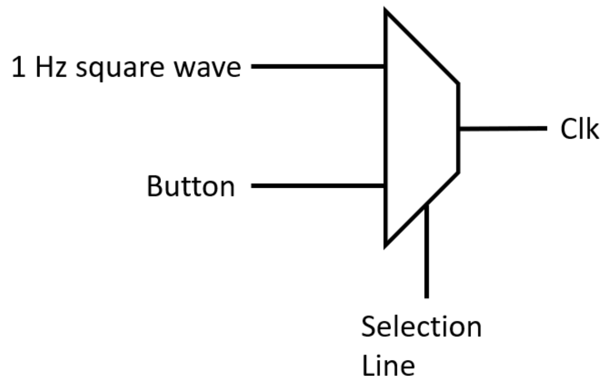


Using Mod-9 counter we can count from 0 to 9 to count the least significant bit (LSB) seconds as well as minutes.

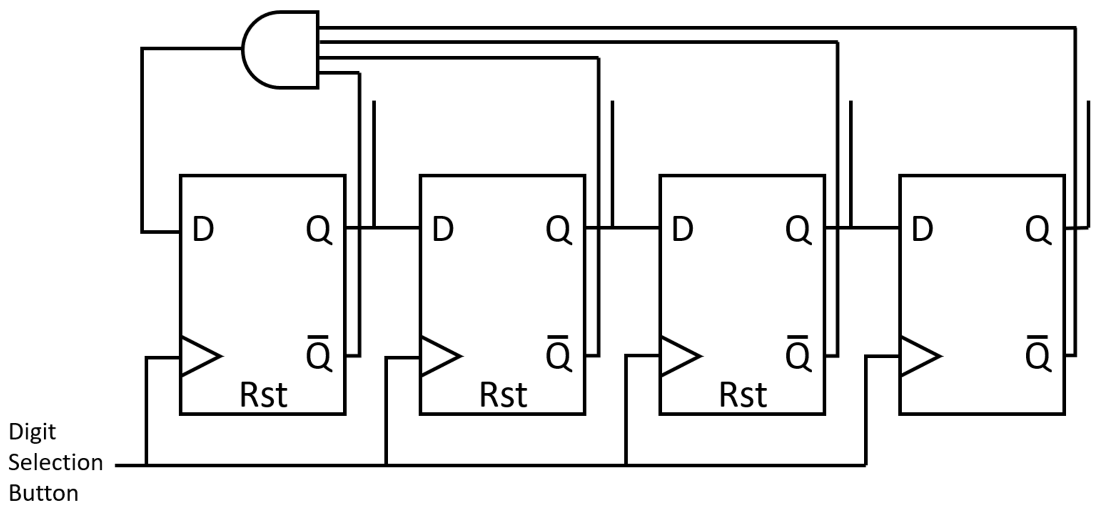


The above circuit shown is counting Hours, Minutes and Seconds from left to right. 2 leftmost is used for counting Hours and last two is used for Seconds counting and left is for Minutes counting.

Using Multiplexer (MUX) as shown above, we are bypassing the FPGA clock and operating with the help of buttons of FPGA to change the time.

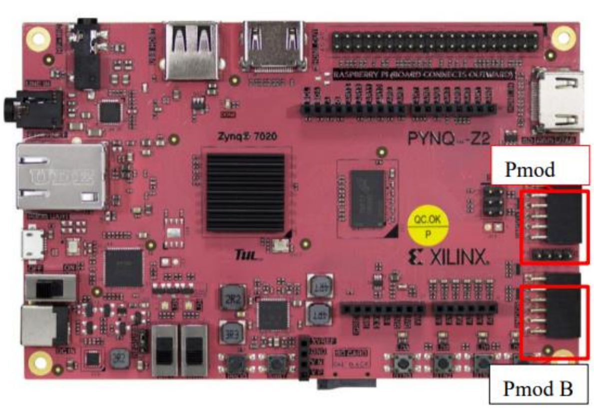


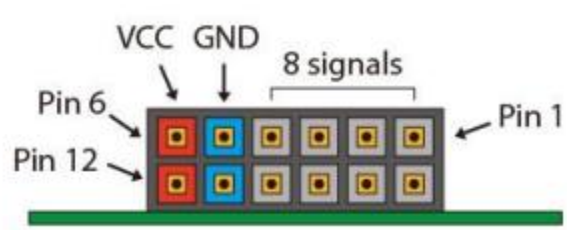
Using Ring Counter, we select the clock's digits that we will change.



Use of FPGA:

We used the Pmod A





Verilog code:

```
`timescale 1ns / 1ps
// MAIN MODULE
module clock(
    input clk,
    input settime,
    input timetoset,
    input selector,
    input rst,
    output [3:0]E,
    output [6:0]O
    );
    wire [3:0]I1,I3;
    wire [2:0]I2,I4;
    wire [1:0]I5;
    wire I6;
    wire [5:0]w1;
    wire w2,clk1,clk2;
    wire r1,r2,r3,r4,r5,r6;

    clockonesec cl(clk,rst,clk1);//CONVERTING DEFAULT CLOCK TO 1 SEC
    selector s1(selector,settime,w1);
    mux4X1 m1(0,timetoset,clk1,clk1,settime,w1[0],r1);
    counter9 c1(r1,rst,I1);
    mux2X1 m2((I1[3]&I1[1]),timetoset,w1[1],r2);
    counter5 c2(r2,rst,I2);
    mux2X1 m3((I2[2]&I2[1]),timetoset,w1[2],r3);
    counter9 c3(r3,rst,I3);
    mux2X1 m4((I3[3]&I3[1]),timetoset,w1[3],r4);
    counter5 c4(r4,rst,I4);
    mux2X1 m5((I4[2]&I4[1]),timetoset,w1[4],r5);
    DFF d2(w3,r5,rst,rst,I5,w3);
    //counter2 c5(r5,rst,I5);
    mux2X1 m6(I5,timetoset,w1[5],r6);
    DFF d1(w2,r6,rst,rst,I6,w2);

    clockdis  cl1(clk,rst,clk2);
    display dis1(clk2,rst,I1,I2,I3,I4,E,O);

endmodule

//MOD-9 COUNTER
module counter9(
    input clk,
    input rst,
    output [3:0]o
    );
```

```
        wire w1,w2,w3,w4;
        DFF DFF1(w1,clk,rst,(o[1]&o[3]),o[0],w1);
        DFF DFF2(w2,w1,rst,(o[1]&o[3]),o[1],w2);
        DFF DFF3(w3,w2,rst,(o[1]&o[3]),o[2],w3);
        DFF DFF4(w4,w3,rst,(o[1]&o[3]),o[3],w4);
endmodule

//MOD-5 COUNTER
module counter5(
    input clk,
    input rst,
    output[2:0]o
    );
        wire w1,w2,w3;
        DFF DFF1(w1,clk,rst,(o[2]&o[1]),o[0],w1);
        DFF DFF2(w2,w1,rst,(o[2]&o[1]),o[1],w2);
        DFF DFF3(w3,w2,rst,(o[2]&o[1]),o[2],w3);

endmodule

//MOD-2 COUNTER
module counter2(
    input clk,
    input rst,
    output [1:0]o
    );
        wire w1,w2;
        DFF DFF1(w1,clk,rst,(o[0]&o[1]),o[0],w1);
        DFF DFF2(w2,w1,rst,(o[0]&o[1]),o[1],w2);

endmodule

//MOD-1 COUNTER
module DFF(D,clk,rst,reset,Q,Qbar);
    input D,clk,rst,reset;
    output reg Q;
    output Qbar;
    assign Qbar=~Q;
    always @(posedge clk or posedge reset or posedge rst ) begin
        if(rst|reset)
            begin
            Q <= 1'b0;
            end
        else
            begin
            Q <= D;
            end
    end
```

```
endmodule

//4:1 MULTIPLEXER
module mux4X1(
    input I0,
    input I1,
    input I2,
    input I3,
    input S0,
    input S1,
    output reg o
    );
    always@(*)
    begin
    case({S0,S1})
        2'b00: o<=I0;
        2'b01: o<=I1;
        2'b10: o<=I2;
        2'b11: o<=I3;
     endcase
     end

endmodule

//2:1 MULTIPLEXER
module mux2X1(
    input I0,
    input I1,
    input S,
    output reg O
    );
    always@(*)
    begin
    case(S)
        0: O <= I0;
        1: O <= I1;
    endcase
    end

endmodule

//REGISTER FOR SELECTING THE DIGIT TO CHANGE
module selector(
    input selectline,
    input rst,
    output [5:0]o
    );
        wire w1,w2,w3,w4,w5,w6;
```

```
        DFF DFF1((w1&w2&w3&w4&w5),selectline,rst,rst,o[0],w1);
        DFF DFF2(o[0],selectline,rst,rst,o[1],w2);
        DFF DFF3(o[1],selectline,rst,rst,o[2],w3);
        DFF DFF4(o[2],selectline,rst,rst,o[3],w4);
        DFF DFF5(o[3],selectline,rst,rst,o[4],w5);
        DFF DFF6(o[4],selectline,rst,rst,o[5],w6);

endmodule

//CONVERTING THE BCD TO 7-SEGMENT DISPLAY OUTPUT FOR MOD-9 COUNTER
module bcdtosevseg9(
    input [3:0]I,
    output reg [6:0]O
    );
    always@(*)
    begin
    case(I)
        4'b0000: O<=7'b0000001;
        4'b0001: O<=7'b1001111;
        4'b0010: O<=7'b0010010;
        4'b0011: O<=7'b0000110;
        4'b0100: O<=7'b1001100;
        4'b0101: O<=7'b0100100;
        4'b0110: O<=7'b0100000;
        4'b0111: O<=7'b0001111;
        4'b1000: O<=7'b0000000;
        4'b1001: O<=7'b0000100;
        Default  O<=7'b1111111;
    endcase
    end
endmodule

//CONVERTING THE BCD TO 7-SEGMENT DISPLAY OUTPUT FOR MOD-5 COUNTER
module bcdtosevseg6(
    input [2:0]I,
    output reg [6:0]O
    );
    always@(*)
    begin
    case(I)
        3'b000: O<=7'b0000001;
        3'b001: O<=7'b1001111;
        3'b010: O<=7'b0010010;
        3'b011: O<=7'b0000110;
        3'b100: O<=7'b1001100;
        3'b101: O<=7'b0100100;
        3'b110: O<=7'b0100000;
```

```
        default O<=7'b1111111;
    endcase
    end
endmodule

//CONVERTING THE BCD TO 7-SEGMENT DISPLAY OUTPUT FOR MOD-2 COUNTER
module bcdtosevseg2(
    input I,
    output reg [6:0]O
    );
    always@(*)
    begin
    case(I)
        1'b00: O<=7'b1001111;
        1'b01: O<=7'b0010010;

        default O<=7'b1111111;
    endcase
    end
endmodule

//CONVERTING THE BCD TO 7-SEGMENT DISPLAY OUTPUT FOR MOD-1 COUNTER
module bcdtosevseg1(
    input I,
    output reg [6:0]O
    );
    always@(*)
    begin
    case(I)
        1'b0: O<=7'b0000001;
        1'b1: O<=7'b1001111;
        default O<=7'b1111111;
    endcase
    end
endmodule

//CONVERTING THE DEFAULT CLOCK OF FPGA TO 1 SEC.
module clockonesec(
    input clk,
    input rst,
    output reg clk1
    );
    reg [25:0]i;
    always@(posedge clk)
    begin
    if(rst)
    begin
    clk1<=1'b0;
```

```
    i=26'b00000000000000000000000001;
    end
    else if(i==26'b11101110011010110010100000)
    begin
    clk1=~clk1;
    i=26'b00000000000000000000000001;
    end
    else
    i=i+26'b00000000000000000000000001;
    end
endmodule

//CREATING CLOCK USED TO DISPLAY ON THE 7-SEGMENT DISPLAY
module clockdis(
    input clk,
    input rst,
    output reg clk1
    );
    reg [16:0]i;
    always@(posedge clk)
    begin

    if(rst)
    begin
    clk1<=1'b0;
    i=17'b00000000000000000;
    end
    else if(i==17'b11000011010100000)
    begin
    clk1=~clk1;
    i=17'b00000000000000000;
    end
    else
    begin
    i=i+17'b00000000000000001;
    end

    end
endmodule

//MODULE USED TO DISPLAY ON THE 7-SEGMENT DISPLAY
module display(
    input clk,
    input rst,
    input [3:0]I1,
    input [2:0]I2,
    input [3:0]I3,
    input [2:0]I4,
```

```
    output reg [3:0]E,
    output reg [6:0]O
    );
    wire [6:0]O1,O2,O3,O4;
    bcdtosevseg9 b1(I1,O1);
    bcdtosevseg6 b2(I2,O2);
    bcdtosevseg9 b3(I3,O3);
    bcdtosevseg6 b4(I4,O4);
    integer i;
    always@(posedge clk or posedge rst)
    begin
    if (rst)
    begin
    i=0;
    E<=4'b0000;
    end
    else if(i==1)
    begin
    E<=4'b1000;
    O<=O1;
    end
    else if (i==2)
    begin
    E<=4'b0100;
    O<=O2;
    end
    else if (i==3)
    begin
    E<=4'b0010;
    O<=O3;
    end
    else if (i==4)
    begin
    E<=4'b0001;
    O<=O4;
    i=0;
    end
    else
    begin
    end
    i=i+1;
    end

endmodule
```

Testbench Code:

```
`timescale 1ns / 1ps

module tb(

    );
    reg clk,settime,timetoset,selector,rst;

    wire [6:0]O;
    wire [3:0]E;
    clock dut(clk,settime,timetoset,selector,rst,E,O);

    initial
    begin
    clk=0;
    settime=1;
    timetoset=0;
    selector=0;
    rst=1;
    #20 rst =0;
    #700 settime=0;
    #5 selector = 1;
    #5 selector = 0;
    #5 timetoset=1;
    #5 timetoset=0;
    #5 timetoset=1;
    #5 timetoset=0;
    #5 selector = 1;
    #5 selector = 0;
    #5 timetoset=1;
    #5 timetoset=0;
    #10 settime=1;
    end
   initial
    begin
     forever #4 clk=~clk;
    end
endmodule
```

Constrain File:

```
#create_clock -period 8.000 -name clk -waveform {0.000 4.000} -add
[get_ports clk]

set_property IOSTANDARD LVCMOS33 [get_ports clk]
set_property IOSTANDARD LVCMOS33 [get_ports rst]
set_property IOSTANDARD LVCMOS33 [get_ports settime]
set_property IOSTANDARD LVCMOS33 [get_ports selector]
```

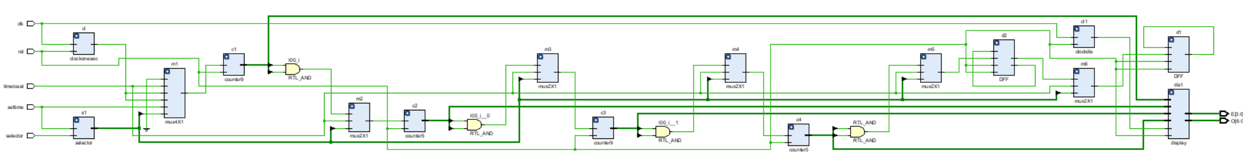
```
set_property IOSTANDARD LVCMOS33 [get_ports timetoset]
set_property IOSTANDARD LVCMOS33 [get_ports {E[3]}]
set_property IOSTANDARD LVCMOS33 [get_ports {O[3]}]
set_property IOSTANDARD LVCMOS33 [get_ports {O[4]}]
set_property IOSTANDARD LVCMOS33 [get_ports {O[5]}]
set_property IOSTANDARD LVCMOS33 [get_ports {O[6]}]
set_property IOSTANDARD LVCMOS33 [get_ports {E[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports {E[2]}]
set_property IOSTANDARD LVCMOS33 [get_ports {E[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {O[2]}]
set_property IOSTANDARD LVCMOS33 [get_ports {O[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {O[0]}]
set_property PACKAGE_PIN H16 [get_ports clk]
set_property PACKAGE_PIN M19 [get_ports rst]
set_property PACKAGE_PIN D19 [get_ports selector]
set_property PACKAGE_PIN M20 [get_ports settime]
set_property PACKAGE_PIN D20 [get_ports timetoset]
set_property PACKAGE_PIN Y18 [get_ports {O[0]}]
set_property PACKAGE_PIN Y19 [get_ports {O[1]}]
set_property PACKAGE_PIN Y16 [get_ports {O[2]}]
set_property PACKAGE_PIN Y17 [get_ports {O[3]}]
set_property PACKAGE_PIN U18 [get_ports {O[4]}]
set_property PACKAGE_PIN U19 [get_ports {O[5]}]
set_property PACKAGE_PIN W18 [get_ports {O[6]}]
set_property PACKAGE_PIN W14 [get_ports {E[0]}]
set_property PACKAGE_PIN Y14 [get_ports {E[1]}]
set_property PACKAGE_PIN T11 [get_ports {E[2]}]
set_property PACKAGE_PIN T10 [get_ports {E[3]}]

set_property CLOCK_DEDICATED_ROUTE FALSE [get_nets selector_IBUF]
```

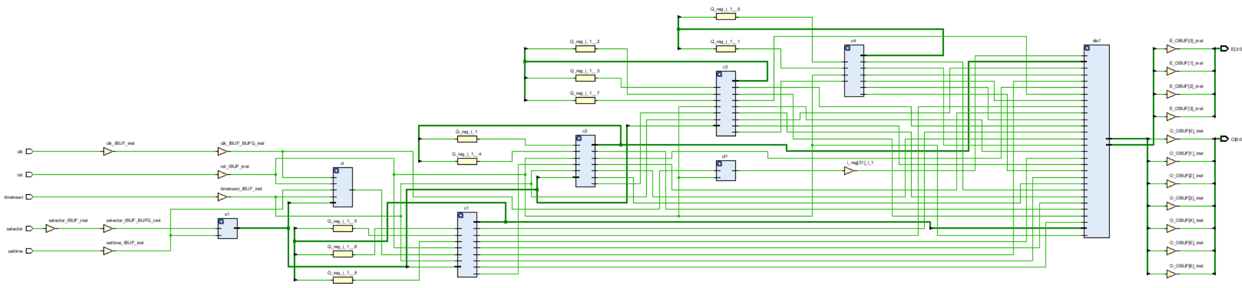
Find Results

Name	Direction	Board Part Pin	Board Part Interface	Interface	Neg Diff Pair	Package Pin	Fixed	Bank	I/O Std	Vcco	Vref	Drive Strength	Slew Type	Pull Type	Off-Chip Termination	IN
clk	IN					H16	✓	35	LVCMOS33*	3.300				NONE	NONE	
rst	IN					M19	✓	35	LVCMOS33*	3.300				NONE	NONE	
selector	IN					D19	✓	35	LVCMOS33*	3.300				NONE	NONE	
settime	IN					M20	✓	35	LVCMOS33*	3.300				NONE	NONE	
timetoset	IN					D20	✓	35	LVCMOS33*	3.300				NONE	NONE	
E[3]	OUT					T10	✓	34	LVCMOS33*	3.300		12		NONE	FP_VTT_50	
E[2]	OUT					T11	✓	34	LVCMOS33*	3.300		12		NONE	FP_VTT_50	
E[1]	OUT					Y14	✓	34	LVCMOS33*	3.300		12		NONE	FP_VTT_50	
E[0]	OUT					W14	✓	34	LVCMOS33*	3.300		12		NONE	FP_VTT_50	
O[6]	OUT					W18	✓	34	LVCMOS33*	3.300		12		NONE	FP_VTT_50	
O[5]	OUT					U19	✓	34	LVCMOS33*	3.300		12		NONE	FP_VTT_50	
O[4]	OUT					U18	✓	34	LVCMOS33*	3.300		12		NONE	FP_VTT_50	
O[3]	OUT					Y17	✓	34	LVCMOS33*	3.300		12		NONE	FP_VTT_50	
O[2]	OUT					Y16	✓	34	LVCMOS33*	3.300		12		NONE	FP_VTT_50	
O[1]	OUT					Y19	✓	34	LVCMOS33*	3.300		12		NONE	FP_VTT_50	
O[0]	OUT					Y18	✓	34	LVCMOS33*	3.300		12		NONE	FP_VTT_50	

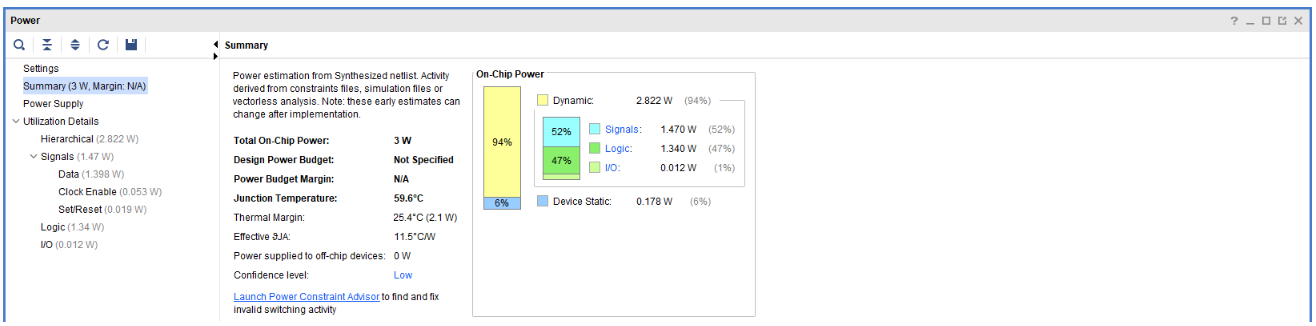
Schematic:



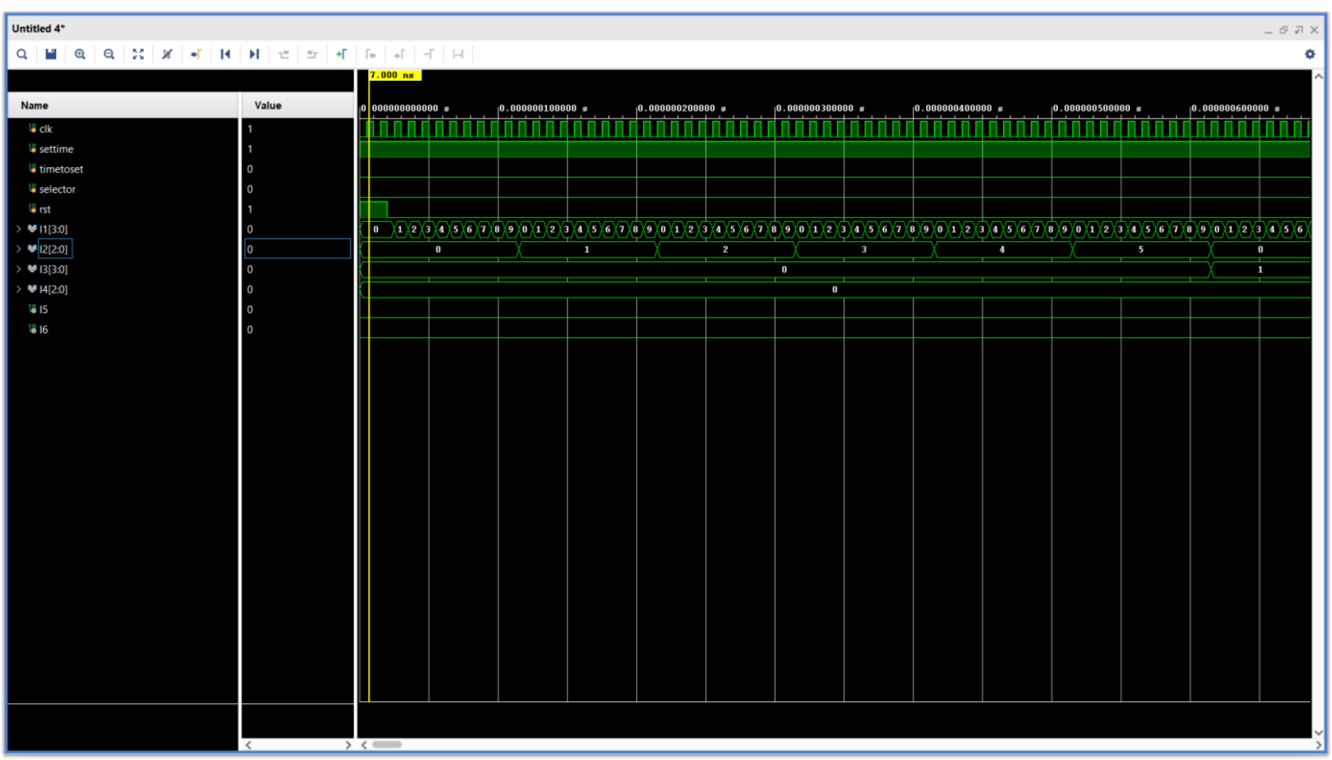
Implemented Schematic:

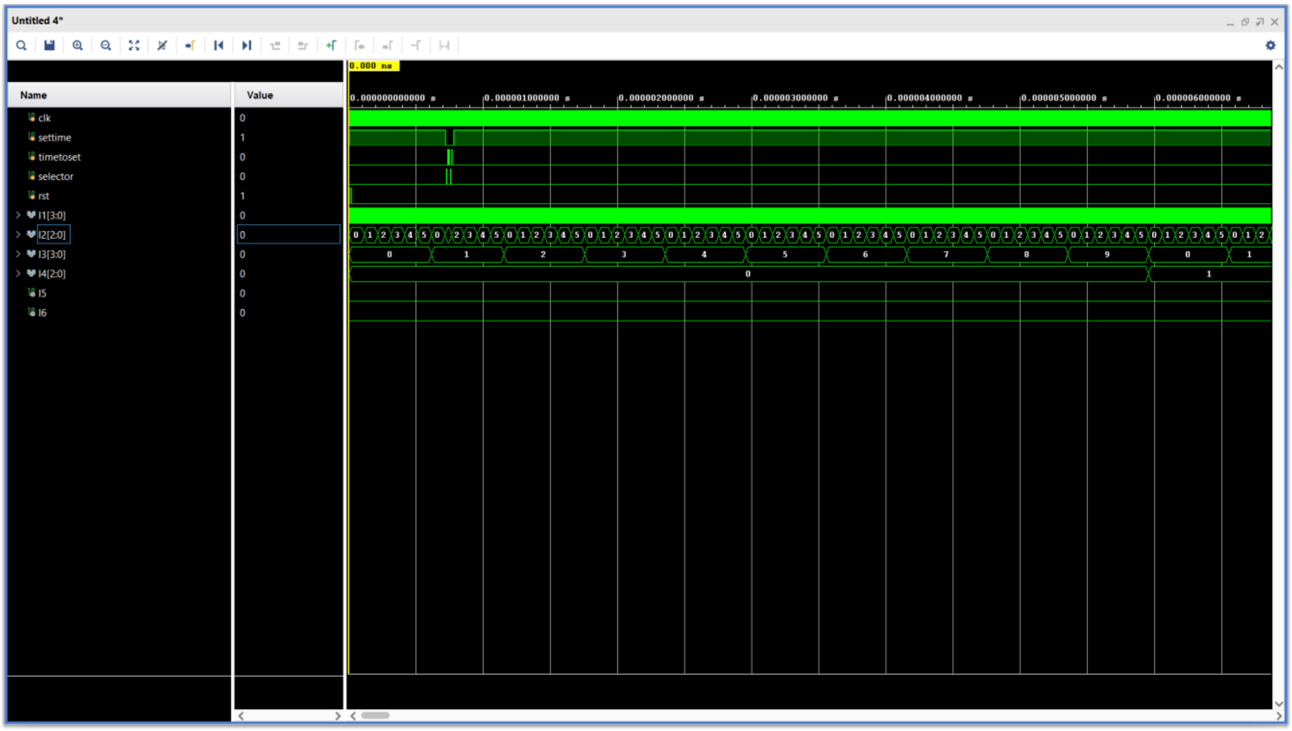


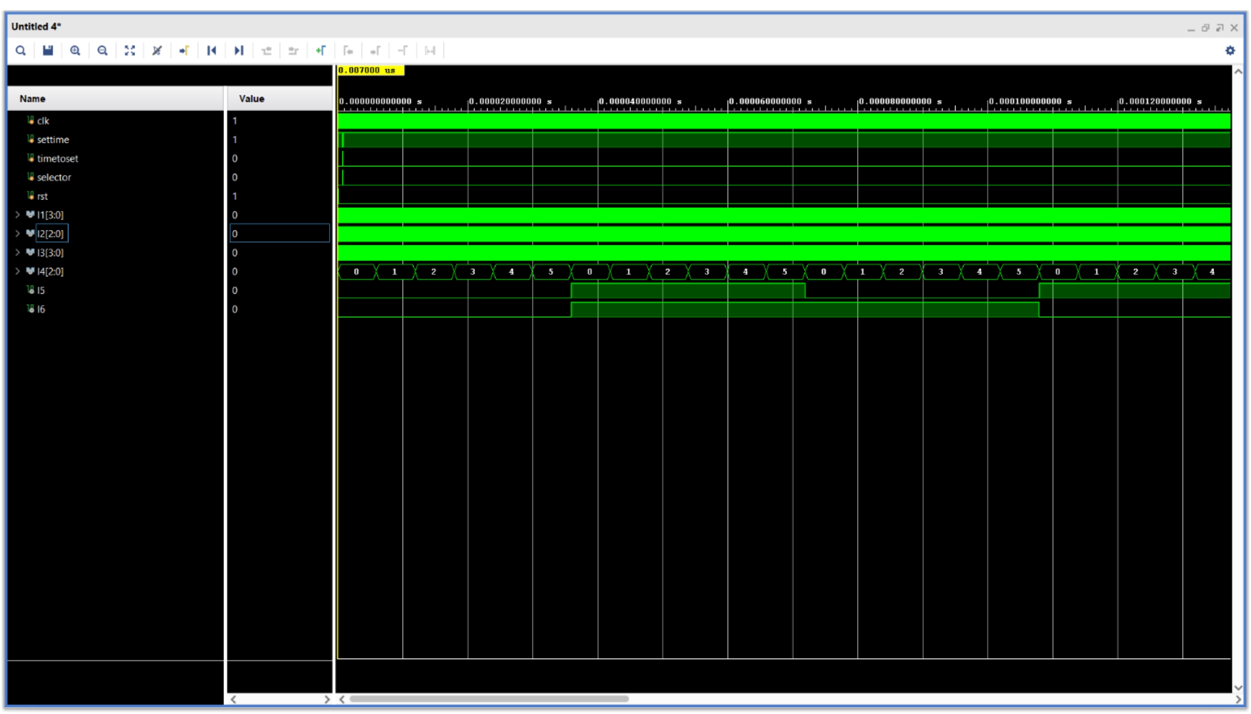
Power Analysis:



Simulations:




Untitled 4*
Name
Value
clk
settime
timetoset
selector
rst
I1[3:0]
I2[2:0]
I3[3:0]
I4[2:0]
I5
I6


Untitled 4*
Name
Value
clk
settime
timetoset
selector
rst
I1[3:0]
I2[2:0]
I3[3:0]
I4[2:0]
I5
I6

Working:

Video Link: Demonstration Link